

Quantitative Investing

PROBLEM SET 4

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Contents. Parts I–III test the CAPM in the time-series framework with the Gibbons, Ross and Shanken (GRS) test, on the 30 value-weight industry portfolios, the 10 past return portfolios and the 25 size and BE/ME portfolios. Part III also examines Roll’s critique using in-sample and out-of-sample tangency portfolios as the market proxy. The capstone adds the profitability factor RMW to the market and tests the augmented model on the 25 size and BE/ME portfolios. The capstone Python code is reproduced in its section, in the order of the original notebook.

PART I: 30 VALUE-WEIGHT INDUSTRY PORTFOLIOS

Data and conventions. Returns are monthly percentages. Portfolio returns are converted to excess returns by subtracting RF; RM-RF is already an excess return.

Question (a)

Table 1: Descriptive statistics, 30 value-weight industry portfolios.

Portfolio	Mean	SD	Sharpe	Portfolio	Mean	SD	Sharpe
Food	0.940	4.685	0.143	Carry	1.153	7.577	0.116
Beer	1.135	7.038	0.123	Mines	0.979	7.356	0.096
Smoke	1.167	5.841	0.154	Coal	1.075	10.998	0.073
Games	1.117	8.945	0.095	Oil	1.034	6.423	0.119
Books	0.889	7.165	0.086	Util	0.887	5.435	0.114
Hshld	0.891	5.737	0.108	Telcm	0.812	4.661	0.116
Clths	0.871	6.225	0.096	Servs	1.237	8.693	0.111
Hlth	1.068	5.501	0.145	BusEq	1.237	6.745	0.143
Chems	1.020	6.307	0.119	Paper	0.964	5.861	0.118
Txtls	0.927	7.936	0.083	Trans	0.929	7.068	0.093
Cnstr	1.030	6.973	0.109	Whlsl	0.869	7.247	0.082
Steel	0.997	8.637	0.084	Rtail	1.061	5.951	0.133
FabPr	1.149	7.268	0.121	Meals	1.065	6.433	0.123
ElcEq	1.202	7.700	0.121	Fin	1.010	6.724	0.110
Autos	1.209	8.578	0.109	Other	0.821	6.582	0.084

Notes: Monthly data, July 1926–June 2026. Mean and standard deviation are based on raw portfolio returns, in percent per month. Sharpe ratios are computed using excess returns, $(R_p - R_f)$, divided by their sample standard deviation.

There is no common ordering across industries, but two patterns are visible. Average returns are only weakly positively related to market beta (correlation about 0.30), while Sharpe ratios are negatively related to beta (correlation about -0.52). Only five industries have monthly Sharpe ratios above the market’s 0.131: Smoke, Health, Food, Business Equipment, and Retail.

Question (b)

For each industry portfolio, we estimated

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{Mt} - R_{ft}) + \varepsilon_{it}.$$

The sample is July 1926 through June 2026, with $T = 1200$, $N = 30$, and $K = 1$.

Table 2: GRS test, 30 industry portfolios on RM-RF.

Statistic	Result
GRS F-statistic	1.7270
Degrees of freedom	$F(30, 1169)$
p -value	0.0091
Individually significant alphas at 5%	4 of 30

Because the p -value is below 1%, we reject the joint null that all 30 industry alphas equal zero. The one-factor CAPM does not fully price the industry portfolios.

Table 3: CAPM time-series regressions, 30 industry portfolios.

Portfolio	Alpha	Beta	$t(\alpha)$	$p(\alpha)$	Portfolio	Alpha	Beta	$t(\alpha)$	$p(\alpha)$
Food	0.167	0.723	2.14	0.0328	Carry	0.061	1.182	0.49	0.6262
Beer	0.232	0.911	1.55	0.1211	Mines	0.075	0.911	0.47	0.6418
Smoke	0.464	0.624	3.31	0.0009	Coal	-0.078	1.269	-0.31	0.7594
Games	-0.120	1.390	-0.81	0.4175	Oil	0.152	0.879	1.18	0.2378
Books	-0.152	1.109	-1.28	0.2008	Util	0.090	0.759	0.85	0.3979
Hshld	0.013	0.875	0.13	0.8977	Telcm	0.079	0.666	0.90	0.3708
Clths	0.016	0.842	0.13	0.8996	Servs	0.352	0.883	1.65	0.0991
Hlth	0.223	0.827	2.30	0.0215	BusEq	0.215	1.082	2.07	0.0391
Chems	0.025	1.042	0.28	0.7762	Paper	0.036	0.947	0.40	0.6858
Txtls	-0.146	1.156	-0.99	0.3205	Trans	-0.139	1.148	-1.33	0.1850
Cnstr	-0.065	1.186	-0.74	0.4597	Whsl	-0.164	1.097	-1.30	0.1949
Steel	-0.227	1.372	-1.67	0.0958	Rtail	0.117	0.970	1.33	0.1833
FabPr	0.014	1.245	0.15	0.8780	Meals	0.140	0.943	1.18	0.2375
ElcEq	0.029	1.299	0.29	0.7697	Fin	-0.064	1.156	-0.79	0.4311
Autos	0.038	1.296	0.25	0.7992	Other	-0.167	1.033	-1.57	0.1169

Notes: Alphas are monthly percentage returns. Alpha t -statistics use the OLS standard errors from the corresponding time-series CAPM regression.

AI integration: code generation — GRS F-statistic. *Prompt 1.* “State the GRS F-statistic formula precisely and explain each term.”

The LLM correctly identified the main structure of the GRS statistic, including the joint alpha term, the factor mean-variance adjustment, and the $F(N, T - N - K)$ distribution. However, the LLM used the original GRS normalization, in which the covariance matrices are normalized differently from the convention specified in this assignment. The assignment explicitly requires the residual covariance matrix to use denominator $T - K - 1$, the factor covariance matrix to use denominator $T - 1$, and the GRS statistic to use the scaling $(T - N - K)/N$. We therefore followed the assignment’s stated convention exactly. The difference is a normalization choice and does not change the qualitative conclusion in this sample. For this problem set, we used

$$F_{GRS} = \frac{T - N - K}{N} \frac{\hat{\alpha}' \hat{\Sigma}_{\epsilon}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}_f^{-1} \bar{f}}.$$

Prompt 2. “Using the GRS formula above, write Python code implementing the GRS test for 30 industry portfolios in a one-factor CAPM. Use residual covariance with denominator $T - K - 1$, factor covariance with denominator $T - 1$, and report the GRS F-statistic and p -value.”

The LLM correctly used residual covariance rather than raw-return covariance and correctly recognized the required F-distribution degrees of freedom. However, it retained the original-GRS

normalization by adding an extra finite-sample scaling adjustment. Because this differs from the convention explicitly specified in the assignment, we used the assignment formula exactly. Using the assignment convention, the 30-industry test gives a GRS F-statistic of **1.7270**, a p -value of **0.0091**, and degrees of freedom **F(30, 1169)**.

Validation checks. We performed the three validation checks required in the assignment:

1. **Zero-alpha check:** Setting all alphas equal to zero produced a GRS statistic of exactly zero.
2. **Reorder invariance:** Permuting the order of the 30 test portfolios did not change the GRS statistic.
3. **Scale invariance:** Multiplying all returns by 100 did not change the GRS statistic.

All three checks passed. As an additional cross-check, we verified the one-factor GRS statistic using the equivalent Sharpe-ratio formulation, which reproduced the same result to numerical precision.

Question (c)

The GRS null is

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_{30} = 0.$$

Under the CAPM, expected excess returns should be fully explained by exposure to the market factor, so every time-series intercept should be zero. Equivalently, the market proxy should be mean-variance efficient relative to the test assets.

Intuitively, GRS asks whether the alphas are jointly too large to be attributed to sampling noise. It combines the pricing errors while accounting for their residual variances and cross-sectional residual correlations. Large, precisely estimated pricing errors that are not redundant with one another contribute more strongly to rejection.

The Sharpe-ratio interpretation is equivalent: if the market proxy is mean-variance efficient, adding the test assets should not significantly improve the maximum attainable Sharpe ratio. Rejecting GRS provides evidence that the market proxy lies inside, rather than on, the mean-variance frontier spanned by the factor and test assets.

The time-series regressions also implicitly impose the CAPM beta risk premium. Since the factor is RM-RF, its sample mean is the estimated market risk premium. The CAPM predicts

$$E[R_i - R_f] = \beta_i E[R_M - R_f],$$

so the intercept α_i measures the portfolio's deviation from this pricing relation.

AI integration: null hypothesis and the Sharpe ratio interpretation. *Prompt 1.* “State the null hypothesis of the GRS test precisely. What does rejecting it imply about the CAPM?”

The LLM correctly stated the joint null hypothesis $H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_N = 0$. It also correctly explained that rejecting the null means the CAPM fails to price the test portfolios jointly. The LLM correctly noted that rejection does not imply that every individual alpha must be statistically significant. However, it did not explicitly state the equivalent mean-variance interpretation required in the assignment: under the null, the market proxy is mean-variance efficient with respect to the test assets. The LLM also did not explicitly explain the final point required in the assignment: in the time-series CAPM, the beta risk premium is implicitly set equal to the sample mean of RM-RF. Therefore, the CAPM pricing relation is $E[R_i - R_f] = \beta_i E[R_M - R_f]$, and the intercept α_i measures the deviation from that relation.

Prompt 2. “The GRS test statistic can be interpreted in terms of Sharpe ratios. Specifically, it is related to whether the test assets can improve the Sharpe ratio of the factor portfolio used as the market proxy. Can you explain this interpretation?”

The LLM correctly explained that the GRS test is related to the improvement in the maximum squared Sharpe ratio that becomes possible when the test assets are added to the factor portfolio. It correctly connected $\alpha' \Sigma_\epsilon^{-1} \alpha$ to the mean-variance value of the pricing errors and explained that rejecting GRS means there is statistically significant evidence that adding the test assets can improve the attainable Sharpe ratio. It also correctly connected rejection to the market proxy lying inside, rather than on, the mean-variance frontier spanned by the factor and test assets. The conceptual interpretation was correct. However, the response again used the original-GRS finite-sample normalization rather than the convention specified in this assignment, so we retained the assignment normalization in the implementation and reported results.

Question (d)

Most industry pricing errors are economically small: the average absolute alpha is about 0.13% per month, and 26 of the 30 individual alphas are not significant at the 5% level. However, the alphas are not randomly distributed. The correlation between beta and alpha is approximately -0.67 .

Lower-beta industries such as Smoke ($\beta = 0.624$, $\alpha = 0.464\%$), Food ($\beta = 0.723$, $\alpha = 0.167\%$), and Health ($\beta = 0.827$, $\alpha = 0.223\%$) have positive alphas, while high-beta industries such as Steel ($\beta = 1.372$, $\alpha = -0.227\%$) and Games ($\beta = 1.390$, $\alpha = -0.120\%$) have negative alphas.

Business Equipment is an exception to the simple low-beta pattern: it has $\beta = 1.082$ but still has a significantly positive alpha of 0.215% per month. Overall, however, the negative beta-alpha relationship is consistent with a flatter security market line than the CAPM predicts. One possible explanation is that market beta alone does not capture all dimensions of risk or expected returns across industries. An imperfect market proxy could also contribute to the pricing errors. The data here show that the one-factor CAPM misses the pattern, although they do not by themselves distinguish among these possible explanations.

PART II: 10 PAST RETURN PORTFOLIOS

Question (e)

The momentum sample is January 1927 through June 2026, with $T = 1194$.

Table 4: Descriptive statistics, 10 past return portfolios.

Portfolio	Mean raw return (%/mo)	Raw return SD (%/mo)	Monthly Sharpe (excess)
Loser	0.372	9.769	0.010
2	0.732	8.037	0.057
3	0.793	6.879	0.076
4	0.908	6.296	0.101
5	0.937	5.841	0.114
6	0.984	5.721	0.125
7	1.026	5.394	0.140
8	1.123	5.278	0.161
9	1.171	5.529	0.163
Winner	1.534	6.498	0.194

Notes: Monthly data, January 1927–June 2026. Mean and standard deviation are based on raw portfolio returns. Sharpe ratios are computed using excess returns, $(R_p - R_f)$, divided by their sample standard deviation.

Descriptive statistics. Unlike the industry portfolios, the past-return portfolios show a clear monotonic pattern. Average returns and Sharpe ratios rise strongly from past losers to past winners. The monthly Sharpe ratio rises from about 0.01 for Loser to about 0.19 for Winner.

Table 5: GRS test, 10 past return portfolios on RM-RF.

Statistic	Result
GRS F-statistic	6.5930
Degrees of freedom	$F(10, 1183)$
p -value	5.34×10^{-10}
Individually significant alphas at 5%	6 of 10

CAPM and GRS results. The GRS test strongly rejects the joint null that all ten momentum alphas equal zero. The rejection is substantially stronger than for the industry portfolios.

Table 6: CAPM time-series regressions, 10 past return portfolios.

Portfolio	Alpha (%/mo)	Beta	Alpha t	Alpha p
Loser	−0.977	1.559	−6.43	0.0000
2	−0.462	1.335	−4.17	0.0000
3	−0.286	1.169	−3.31	0.0010
4	−0.120	1.094	−1.68	0.0924
5	−0.048	1.032	−0.80	0.4236
6	0.004	1.026	0.07	0.9440
7	0.090	0.961	1.75	0.0809
8	0.207	0.933	3.87	0.0001
9	0.239	0.957	3.74	0.0002
Winner	0.552	1.028	5.36	0.0000

Notes: Alphas are monthly percentage returns. Alpha t -statistics use the OLS standard errors from the corresponding time-series CAPM regression.

The CAPM has particular difficulty explaining the winner-loser spread. Across the ten portfolios, the correlation between market beta and average excess return is approximately -0.82 , and the correlation between beta and alpha is approximately -0.91 .

The Loser portfolio has the highest beta, $\beta = 1.559$, but the lowest average return and a large negative alpha of -0.977% per month. The Winner portfolio has a much lower beta, $\beta = 1.028$, but a much higher average return and a positive alpha of 0.552% per month.

Thus, market beta points in the wrong direction for explaining the momentum return spread. Past losers tend to have negative alphas and past winners positive alphas. The winner and loser alphas have opposite signs, although the magnitudes are not symmetric because the loser-side pricing error is larger.

One interpretation is that momentum captures exposure to an omitted priced risk factor that is not included in the CAPM. Another is behavioral underreaction or slow information diffusion, which can create predictable continuation in returns. The CAPM itself cannot distinguish between these explanations.

AI integration: momentum and the CAPM. *Prompt.* “The CAPM typically rejects more strongly for portfolios sorted on past returns than for industry portfolios. Why is momentum particularly difficult for the CAPM to price? Is momentum evidence against the CAPM, against efficient markets, or both?”

The LLM correctly explained that the CAPM has no direct mechanism for momentum: market beta cannot explain why past winners subsequently earn much higher returns than past losers. It also correctly distinguished two broad interpretations:

- **Risk-based explanation:** momentum may reflect exposure to an omitted priced factor, so the CAPM is incomplete while market efficiency could still hold.

- **Behavioral explanation:** underreaction, slow information diffusion, or other investor behavior may create persistent mispricing.

The LLM therefore correctly concluded that momentum is strong evidence against the CAPM, but is not automatically evidence against market efficiency because of the joint-hypothesis problem.

However, part of its numerical intuition was too generic. It suggested that winner and loser portfolios often have similar market betas. This does not match our sample. In our data, Loser beta = **1.559**, Winner beta = **1.028**, the correlation between beta and average excess return across the ten portfolios is approximately **-0.82**, and the correlation between beta and alpha across the ten portfolios is approximately **-0.91**. Thus, higher market beta is actually associated with lower average returns and more negative alphas across the momentum portfolios.

The alpha pattern does match the LLM's general prediction: Loser alpha = **-0.977% per month** and Winner alpha = **+0.552% per month**. The winner and loser alphas therefore have opposite signs, although their magnitudes are not symmetric because the loser-side pricing error is larger. The LLM correctly distinguished the risk-based and behavioral explanations, but its discussion was mostly conceptual and did not provide much empirical evidence that would distinguish between the two.

Bonus: time-series vs. cross-sectional momentum. *Prompt.* “Are time-series momentum (TSMOM) and cross-sectional momentum (UMD) the same thing? Explain the difference.”

The LLM correctly explained that the two strategies are related but not identical. Cross-sectional momentum ranks assets relative to one another and goes long relative winners and short relative losers. UMD is an example of this type of momentum strategy. Time-series momentum instead uses the sign of each asset's own past return: an asset with positive past performance is typically held long, while an asset with negative past performance is held short. Therefore, cross-sectional momentum uses relative performance across assets, while time-series momentum uses each asset's own return history. The two strategies can even take opposite positions in the same asset.

PART III: 25 SIZE AND BE/ME PORTFOLIOS

Sample. 1926-07 to 2026-06, $T = 1200$, $N = 25$. Returns are in excess of the one-month T-bill rate, in percent per month (Parts I and II report raw means; Sharpe ratios use excess returns throughout). Rows sort on size (S = small, B = big); columns sort on BE/ME (L = low, H = high).

Question (f), repeating Part I(a)**Table 7:** Summary statistics, 25 size and BE/ME portfolios, percent per month.

	Mean excess return					Standard deviation				
	L	2	3	4	H	L	2	3	4	H
S	0.58	0.68	1.00	1.14	1.35	11.88	9.62	8.82	8.25	9.13
2	0.66	0.95	0.98	1.04	1.24	7.96	7.46	7.21	7.41	8.64
3	0.73	0.91	0.92	1.01	1.11	7.34	6.47	6.45	6.91	8.34
4	0.73	0.79	0.86	0.97	1.03	6.17	6.08	6.28	6.80	8.52
B	0.69	0.66	0.73	0.67	1.01	5.34	5.24	5.56	6.50	8.48
Sharpe ratio										
S	0.049	0.071	0.113	0.138	0.148					
2	0.083	0.127	0.137	0.141	0.144					
3	0.100	0.141	0.142	0.145	0.133					
4	0.118	0.129	0.136	0.143	0.121					
B	0.130	0.125	0.131	0.103	0.119					

Monthly Sharpe ratio = $\bar{R}_i^e / \hat{\sigma}_i$. Market proxy (RM−RF): mean 0.70, standard deviation 5.31, Sharpe ratio 0.131.

Discussion.

- **Is there a discernible pattern in the average returns or Sharpe ratios?** The dominant pattern is a value effect. Average returns and Sharpe ratios rise with BE/ME in every size quintile, and the spread is steepest among small stocks. The size effect is conditional: small value beats big value, but small growth earns less than big growth and is the worst of the 25 portfolios, with the lowest mean and the highest volatility. Unlike the industries in Part I, which span a similar range of Sharpe ratios with no ordering, the dispersion here lines up with the characteristic we sorted on, which is exactly what a single beta will struggle to match.

Question (f), repeating Part I(b)

For each portfolio we estimate

$$R_{it} - R_{ft} = \alpha_i + \beta_{iM} (R_{Mt} - R_{ft}) + e_{it},$$

and test $H_0 : \alpha_1 = \dots = \alpha_{25} = 0$ with the GRS statistic, distributed $F(N, T - N - K) = F(25, 1174)$ under the null.

Table 8: CAPM time-series regressions, 25 size and BE/ME portfolios. Intercepts in percent per month.

	$\hat{\alpha}_i$					$t(\hat{\alpha}_i)$				
	L	2	3	4	H	L	2	3	4	H
S	−0.53	−0.29	0.06	0.26	0.41	−2.20	−1.59	0.38	1.88	2.48
2	−0.22	0.09	0.15	0.20	0.29	−1.80	0.88	1.53	1.88	2.12
3	−0.13	0.13	0.14	0.19	0.17	−1.43	1.77	1.88	2.17	1.37
4	−0.03	0.03	0.09	0.17	0.07	−0.41	0.50	1.36	1.96	0.57
B	0.02	0.00	0.06	−0.09	0.10	0.51	0.01	0.90	−1.05	0.72

	$\hat{\beta}_{iM}$				
S	1.60	1.39	1.36	1.26	1.36
2	1.27	1.23	1.20	1.21	1.37
3	1.25	1.13	1.12	1.17	1.35
4	1.09	1.09	1.10	1.16	1.39
B	0.96	0.94	0.96	1.09	1.30

GRS = 3.37, $p < 0.001$, $F(25, 1174)$. $\hat{\Sigma}$ uses denominator $T - K - 1$; the factor variance uses $T - 1$.

- **What do the GRS F-statistic and its p -value show?** The GRS test rejects ($F = 3.37$, $p < 0.001$), so the market proxy is not mean–variance efficient with respect to the 25 size and BE/ME portfolios.

Question (f), repeating Part I(d)

Discussion.

- **What are the sign and magnitude of the intercepts?** The intercepts are negative for growth and positive for value. The spread is widest among small stocks, from about -0.5% per month for small growth to $+0.4\%$ for small value. Among the largest stocks the intercepts are close to zero.
- **Does the CAPM have difficulty pricing any portfolios in particular?** The CAPM has most difficulty with small growth and small value. Small growth is overpriced and small value underpriced relative to their betas, while big stocks are priced reasonably well. Only five intercepts are significant on their own, but the residuals are strongly correlated across portfolios, so the GRS test reads the alphas as one systematic pattern rather than 25 small deviations.
- **Why?** Beta cannot produce the spread, which points to a missing factor for value and mispricing for small growth. Within a size quintile beta barely changes across BE/ME, and small growth has the highest beta of all 25 portfolios despite the lowest return. Additional checks (not tabulated) show that the value spread moves as a common component of the CAPM residuals but does not lose money in market crashes, so it is not compensation for market risk. Small growth has the highest skewness and idiosyncratic volatility of the 25 portfolios, a negative alpha in every subsample, and stays mispriced even after adding a value factor, which fits investors overpaying for lottery-like stocks. In both cases, the pattern lies outside what market beta can explain.

Question (g)

We replace RM–RF with the in-sample tangency portfolio of the 25 portfolios, with weights

$$w_T = \frac{\hat{\Sigma}^{-1} \hat{\mu}}{\mathbf{1}' \hat{\Sigma}^{-1} \hat{\mu}}$$

estimated over the full sample, and rerun the 25 time-series regressions and the GRS test.

Table 9: GRS test with the in-sample tangency portfolio of the 25 size and BE/ME portfolios as the market proxy, July 1926 – June 2026.

	RM–RF	In-sample tangency
GRS F -statistic	3.37	0.00
p -value	< 0.001	1.00
Largest $ \hat{\alpha}_i $ (% per month)	0.53	0.00
Proxy Sharpe ratio (monthly)	0.131	0.299

Tangency weights sum to one, with gross exposure 13.86 and individual weights between -1.44 and 1.06 . All 25 intercepts are below 10^{-14} .

Discussion.

- **Does the GRS test reject or fail to reject when we use the in-sample tangency portfolio as the market proxy?** The GRS test fails to reject. The statistic is zero and the p -value is one: every intercept is zero up to rounding error, so the in-sample tangency portfolio prices all 25 portfolios perfectly.
- **Is this result surprising?** The result is not surprising. The in-sample tangency portfolio is built from the same sample moments used in the test, so it gives every portfolio a zero intercept by construction. Since $w_T \propto \hat{\Sigma}^{-1}\hat{\mu}$, each portfolio's covariance with the tangency portfolio is proportional to its mean, so $\hat{\mu}_i = \hat{\beta}_{iT}\hat{\mu}_T$ and $\hat{\alpha}_i = \hat{\mu}_i - \hat{\beta}_{iT}\hat{\mu}_T = 0$ for every i . In Sharpe-ratio terms, the proxy is already the maximum-Sharpe-ratio combination of the test assets, so adding them cannot improve it. The result therefore says nothing about the CAPM. The proxy is a levered long-short portfolio of the test assets, not a market portfolio, and it illustrates Roll's critique just as the perfect cross-sectional fit did in Problem Set 3, Question (2).

Question (h)

Sample 1 contains odd-numbered months in even years and even-numbered months in odd years; Sample 2 contains the rest, giving 600 months each. We estimate tangency weights from Sample 1 moments and apply them to Sample 2 returns, then reverse the roles. The combined, chronologically sorted series is the out-of-sample tangency portfolio, which we use as the market proxy in the 25 time-series regressions.

Table 10: GRS tests on the 25 size and BE/ME portfolios under three market proxies, July 1926 – June 2026.

	RM–RF	In-sample tangency	Out-of-sample tangency
GRS F -statistic	3.37	0.00	2.78
p -value	< 0.001	1.00	< 0.001
Proxy Sharpe ratio (monthly)	0.131	0.299	0.216

Out-of-sample weights sum to one in each half, with gross exposure 14.81 (estimated on Sample 1) and 18.44 (Sample 2). The correlation between the two weight vectors is 0.49. All 25 intercepts are positive, between 0.14 and 0.46% per month.

Discussion.

- **Does the GRS test reject or fail to reject when we use the out-of-sample tangency portfolio as the market proxy?** The GRS test rejects. The out-of-sample tangency portfolio leaves significant pricing errors in the 25 portfolios.

- **Why?** It rejects because the weights are estimated with error. The identity in (g) holds only when the weights and the test use the same sample moments. Here they do not, and the mean estimates are noisy enough that the two halves produce quite different weights. A portfolio that is efficient in its own half loses about a third of its Sharpe ratio in the other, so it lies inside the frontier of the test assets. Because the proxy is noisy, the 25 portfolios have low betas on it, so their beta-implied returns fall short of their average returns and all 25 intercepts are positive. The rejection therefore reflects the noisy proxy as much as the pricing model.
- **Is this result surprising?** The result is not surprising. Optimized weights fit the sample they come from and travel poorly, so once the weights come from a different sample, nothing forces the intercepts to zero and the test regains its power. The portfolio still beats the market's Sharpe ratio and lowers the GRS statistic, but the test asks whether the proxy is efficient, not whether it beats the market, and out of sample it is not.

AI integration: Roll's critique. Before comparing (g) and (h), we asked Claude Sonnet 5.5 (chat mode) why the in-sample tangency portfolio makes the GRS test fail to reject trivially, and why the out-of-sample version does not. It answered that GRS is a monotone function of $(SR^{*2} - SR_f^2)/(1 + SR_f^2)$, that an in-sample tangency portfolio attains $SR_f = SR^*$ by construction, and that weights estimated on a different sample leave the proxy inside the test-sample frontier, restoring power. It added that estimation error in those weights makes the test over-reject as a test of the underlying model.

- Did the in-sample tangency portfolio in (g) produce a GRS statistic near zero and a very high p -value? Did the LLM's explanation correctly predict this?** Yes, on both counts. The in-sample tangency portfolio produced a GRS statistic of zero and a p -value of one, exactly as the LLM predicted.
- Does the LLM correctly identify that the in-sample tangency portfolio makes all alphas exactly zero, or does it give a vaguer explanation about overfitting?** The LLM identified the exact identity. It showed that $\text{Cov}(R_i, f) = \hat{\mu}_i$, so $\hat{\beta}_i = \hat{\mu}_i/\mu_f$ and every $\hat{\alpha}_i$ is zero, rather than appealing loosely to overfitting.
- Does the out-of-sample tangency portfolio in (h) restore the GRS test's power? How does this reinforce the Roll critique?** Yes, and the contrast reinforces Roll's critique. With the same assets, the same data and the same model, the verdict moves from a perfect fit to a clear rejection depending only on how the proxy is built. The LLM's own caveat makes the same point without naming Roll: the out-of-sample rejection partly reflects estimation error in the proxy, so neither result is a clean verdict on the CAPM. The LLM did overstate one point: it claimed the exact F distribution applies out of sample because the proxy is a given regressor. In our construction each half's proxy uses weights estimated from the other half's test-asset returns, so the F distribution is only approximate. GRS can only ever test whether the chosen proxy is efficient.

Question (i)

We replace RM–RF with the in-sample tangency portfolio of the 30 value-weight industry portfolios, estimated over the full sample, and rerun the GRS test on the 25 size and BE/ME portfolios.

Table 11: CAPM intercepts with the in-sample industry tangency portfolio as the market proxy, percent per month.

	$\hat{\alpha}_i$					$t(\hat{\alpha}_i)$				
	L	2	3	4	H	L	2	3	4	H
S	-0.46	-0.10	0.26	0.47	0.62	-1.40	-0.36	1.06	2.03	2.44
2	-0.17	0.19	0.25	0.33	0.46	-0.81	0.95	1.32	1.63	1.96
3	-0.09	0.14	0.19	0.27	0.31	-0.48	0.86	1.14	1.46	1.37
4	-0.10	0.02	0.11	0.22	0.21	-0.64	0.11	0.66	1.21	0.91
B	-0.12	-0.07	0.03	-0.05	0.20	-1.00	-0.54	0.22	-0.31	0.86

GRS = 3.51, $p < 0.001$, against 3.37 with RM–RF. Monthly Sharpe ratio of the industry tangency portfolio: 0.236.

Discussion.

- **How well do we price the 25 size and BE/ME portfolios?** The industry tangency portfolio prices the 25 portfolios no better than the market. The GRS test still rejects, with a slightly larger statistic than under RM–RF, and the intercepts keep the same pattern: negative for small growth, positive and increasing in BE/ME elsewhere.
- **Is this surprising? Why or why not?** This is not surprising. The tangency portfolio is efficient only relative to the assets it is built from. Industries are not sorted on size or BE/ME, so their optimal combination carries little of the value spread that drives the 25 portfolios' returns. A higher Sharpe ratio than the market does not help: what matters is whether the proxy lies on the frontier of the test assets, and the low correlation between the two tangency portfolios in (k) shows that it does not. Efficiency within one set of assets does not carry over to another.

Question (j)

We replace RM–RF with the in-sample tangency portfolio of the 10 past return portfolios, estimated over the full sample, and rerun the GRS test on the 25 size and BE/ME portfolios. Because the past return portfolios begin in January 1927, this test uses 1927-01 to 2026-06 ($T = 1194$).

Table 12: CAPM intercepts with the in-sample past-return tangency portfolio as the market proxy, percent per month, 1927-01 to 2026-06.

	$\hat{\alpha}_i$					$t(\hat{\alpha}_i)$				
	L	2	3	4	H	L	2	3	4	H
S	-0.20	-0.07	0.35	0.50	0.74	-0.58	-0.24	1.39	2.10	2.77
2	-0.17	0.21	0.26	0.38	0.53	-0.77	0.99	1.30	1.83	2.15
3	-0.07	0.12	0.19	0.31	0.39	-0.37	0.71	1.09	1.60	1.65
4	-0.11	0.02	0.12	0.28	0.28	-0.66	0.15	0.71	1.49	1.17
B	-0.07	-0.07	0.07	-0.00	0.28	-0.54	-0.52	0.44	-0.02	1.17

GRS = 3.36, $p < 0.001$, $F(25, 1168)$, against 3.37 with RM–RF over the same sample. The tangency portfolio is mainly long Winner (0.89) and short Loser (−0.96), with a monthly Sharpe ratio of 0.264.

Discussion.

- **How well do we price the 25 size and BE/ME portfolios?** No better than the market. The GRS statistic is essentially unchanged from the CAPM, but the intercepts now rise even more steeply with BE/ME, reaching 0.74% per month for small value, and their average absolute size increases by about half.

- **Is this surprising? Why or why not?** The result is surprising at first, but understandable. We expected the momentum tangency portfolio to do better, since its Sharpe ratio is about twice the market's. As in (i), however, the test asks whether the proxy is efficient relative to the 25 portfolios, and a bet on winners over losers carries no information about value. It even works against value: value portfolios move with past losers, so the proxy expects little from them and their alphas grow. The GRS statistic holds steady only because those larger alphas are also noisier. A better portfolio is not necessarily a better proxy.

Question (k)

We form the in-sample tangency portfolio of each of the three portfolio sets and correlate their monthly returns over the common sample, 1927-01 to 2026-06 ($T = 1194$).

Table 13: Correlation matrix of the in-sample tangency portfolios, 1927-01 to 2026-06.

	30 industries	10 past return	25 size and BE/ME
30 industries	1.00	0.40	0.27
10 past return	0.40	1.00	0.25
25 size and BE/ME	0.27	0.25	1.00

Monthly Sharpe ratios: 0.232, 0.264 and 0.299. Correlations with RM–RF: 0.61, 0.52 and 0.44.

Discussion.

- **Comment on these correlations in light of (i) and (j).** The three tangency portfolios are only weakly correlated, so they are driven by separate factors rather than one beta measured three ways. Industry differences, past winners over losers and value are different sources of return. This explains (i) and (j): a proxy that shares little of the 25 portfolios' movement cannot lie on their frontier. A single factor that prices all three sets would have to span all three directions at once.

CAPSTONE EXERCISE: CHOOSE A FACTOR, BUILD A TEST

Step 1

Choose RMW factor.

Step 2

We use the 25 size and BE/ME portfolios as the test assets and RMW (Robust Minus Weak) as the additional factor. We expect RMW to help price these portfolios because profitability captures a dimension of expected returns that is not included in the CAPM market factor and may vary systematically across firms with different size and book-to-market characteristics.

Before examining the results, our hypothesis is that adding RMW to the market factor will reduce the GRS F-statistic relative to the CAPM, although we do not expect it to eliminate all pricing errors. We expect the largest alpha reductions to occur for portfolios with stronger RMW exposures, while portfolios with weak profitability exposure should show smaller improvements.

Economically, RMW may earn a positive risk premium because firms with weak profitability could be riskier and therefore require different expected returns, or the profitability premium may reflect systematic mispricing if investors do not fully incorporate information about firms' profitability into prices. The test will examine whether adding RMW materially reduces the pricing errors of the 25 portfolios beyond what market beta alone can explain.

Question 3a

The notebook cells are reproduced in order. The GRS validation output follows the validation cell, and the full alpha comparison follows the last cell.

```

1 import pandas as pd
2 import numpy as np
3 import statsmodels.api as sm
4 from scipy.stats import f
5
6 excel_file = "Problem_Set4_2026.xlsx"
7 factor_file = "F-F_Research_Data_5_Factors_2x3.csv"
8 pd.set_option("display.max_columns", 35)
9 pd.set_option("display.width", 200)
10
11 book = pd.ExcelFile(excel_file)
12 print("Excel sheets:", book.sheet_names)
13
14 portfolio_raw = pd.read_excel(
15     book, sheet_name="25 size and BEME portfolios", header=None
16 )
17 market_raw = pd.read_excel(
18     book, sheet_name="Market, Rf", header=None
19 )
20
21 print("\nPortfolio header rows:")
22 print(portfolio_raw.iloc[:3].to_string(index=False, header=False))
23
24 print("\nMarket header rows:")
25 print(market_raw.iloc[:2].to_string(index=False, header=False))
26
27 # Preserve the workbook's original size-first, BE/ME-second order.
28 portfolio_cols = [
29     f"S{s}_BM{b}" for s in range(1, 6) for b in range(1, 6)
30 ]
31
32 portfolios = portfolio_raw.iloc[3:].copy()
33 assert portfolios.shape[1] == 26
34 portfolios.columns = ["Date"] + portfolio_cols
35
36 market = market_raw.iloc[2:].copy()
37 assert market.shape[1] == 3
38 assert market_raw.iloc[1, 1:].tolist() == ["RM-RF", "RF"]
39 market.columns = ["Date", "RM-RF", "RF"]
40
41 # The CSV header is on line 5.
42 ff = pd.read_csv(factor_file, skiprows=4)
43 ff.columns = ff.columns.str.strip()
44 ff = ff.rename(columns={ff.columns[0]: "Date"})
45
46 print("\nCSV columns:", ff.columns.tolist())
47
48 # Monthly dates have six digits. Annual dates have four.
49 date_text = ff["Date"].astype(str).str.strip()
50 monthly_rows = date_text.str.fullmatch(r"\d{6}")
51
52 print("CSV rows outside monthly section:", int((~monthly_rows).sum()))
53
54 # Retain only RMW from the external factor file.
55 rmw = ff.loc[monthly_rows, ["Date", "RMW"]].copy()
56
57 for name, data in [
58     ("Portfolios", portfolios),
59     ("Workbook market and RF", market),
60     ("Monthly RMW", rmw)
61 ]:
62     print("\n" + name)
63     print("Dimensions:", data.shape)

```

```

64     print("Columns:", data.columns.tolist())
65     print(data.head().to_string(index=False))
66     print("Data types:")
67     print(data.dtypes)

1  sources = [
2      ("Portfolios", portfolios),
3      ("Workbook market and RF", market),
4      ("Monthly RMW", rmw)
5  ]
6
7  for name, data in sources:
8      dates = pd.to_numeric(data["Date"], errors="raise")
9
10     assert dates.notna().all(), name + ": missing dates"
11     assert (dates % 1 == 0).all(), name + ": invalid YYYYMM dates"
12
13     data["Date"] = pd.to_datetime(
14         dates.astype(int).astype(str), format="%Y%m", errors="raise"
15     ).dt.to_period("M")
16
17     value_cols = data.columns.drop("Date")
18     data[value_cols] = data[value_cols].apply(
19         pd.to_numeric, errors="raise"
20     )
21
22     # Recognize standard missing-return codes explicitly.
23     sentinel_count = data[value_cols].isin([-99.99, -999]).sum().sum()
24     print(f"\n{name}: missing-value codes = {sentinel_count}")
25
26     data[value_cols] = data[value_cols].replace(
27         [-99.99, -999], np.nan
28     )
29
30     print("Missing values before merging:")
31     print(data[value_cols].isna().sum().to_string())
32     print("Rows with any missing return:",
33         int(data[value_cols].isna().any(axis=1).sum()))
34     print("Duplicate dates:", int(data["Date"].duplicated().sum()))
35     print("Range:", data["Date"].min(), "to", data["Date"].max())
36
37     assert not data["Date"].duplicated().any(), name + ": duplicate dates"
38
39     # Both models will use this single merged sample.
40     common = portfolios.merge(
41         market, on="Date", how="inner", validate="one_to_one"
42     )
43     common = common.merge(
44         rmw, on="Date", how="inner", validate="one_to_one"
45     )
46     common = common.sort_values("Date").set_index("Date")
47
48     for name, data in sources:
49         excluded = data.loc[~data["Date"].isin(common.index), "Date"]
50         print(f"\n{name}: {len(excluded)} months excluded by date overlap")
51         if len(excluded) > 0:
52             print("Excluded range:", excluded.min(), "to", excluded.max())
53
54     print("\nMissing values after merging:")
55     print(common.isna().sum().to_string())
56     print("Rows with any missing value:",
57         int(common.isna().any(axis=1).sum()))
58
59     assert len(common) > 0, "No overlapping observations."
60     assert not common.isna().any().any(), "Missing values remain. Stop here."
61     assert np.isfinite(common.to_numpy()).all(), "Nonfinite values remain."
62
63     expected_months = pd.period_range(

```

```

64     common.index.min(), common.index.max(), freq="M"
65 )
66 assert common.index.equals(expected_months), "There are gaps in the sample."
67
68 print("\nCommon sample:")
69 print("Start:", common.index.min())
70 print("End:", common.index.max())
71 print("T:", len(common))
72 print("Units: percentage points for portfolios, RM-RF, RF, and RMW.")
73 print("\nCleaned data types:")
74 print(common.dtypes)

1  Y = common[portfolio_cols].sub(common["RF"], axis=0)
2
3  F_capm = common[["RM-RF"]].copy()
4  F_aug = common[["RM-RF", "RMW"]].copy()
5
6  assert Y.index.equals(F_capm.index)
7  assert Y.index.equals(F_aug.index)
8  assert Y.shape[1] == 25
9
10 print("Portfolio excess returns = raw portfolio returns - workbook RF.")
11 print("RM-RF and RMW are used directly.")
12 print("All returns remain in percentage points.")
13 print(Y.head().to_string())

1  X_capm = sm.add_constant(F_capm, has_constant="add")
2
3  capm_rows = []
4  capm_residuals = pd.DataFrame(index=Y.index)
5  capm_fitted = pd.DataFrame(index=Y.index)
6
7  for portfolio in portfolio_cols:
8      model = sm.OLS(Y[portfolio], X_capm, missing="raise").fit()
9
10     capm_rows.append({
11         "Portfolio": portfolio,
12         "Alpha": model.params["const"],
13         "Alpha SE": model.bse["const"],
14         "Alpha t-stat": model.tvalues["const"],
15         "Market beta": model.params["RM-RF"],
16         "Market beta t-stat": model.tvalues["RM-RF"]
17     })
18
19     capm_residuals[portfolio] = model.resid
20     capm_fitted[portfolio] = model.fittedvalues
21
22 capm_results = pd.DataFrame(capm_rows).set_index("Portfolio")
23 print(capm_results.to_string(float_format=lambda x: f"{x:.6f}"))

1  def grs_test(alpha, residuals, factors):
2      alpha = np.asarray(alpha, dtype=float).reshape(-1)
3      E = np.asarray(residuals, dtype=float)
4      F = np.asarray(factors, dtype=float)
5
6      assert E.ndim == 2 and F.ndim == 2
7
8      T, N = E.shape
9      K = F.shape[1]
10
11     assert F.shape[0] == T
12     assert len(alpha) == N
13     assert T > N + K
14     assert np.isfinite(alpha).all()
15     assert np.isfinite(E).all()
16     assert np.isfinite(F).all()
17

```

```

18     # Covariance of OLS residuals, not raw portfolio returns.
19     sigma = E.T @ E / (T - K - 1)
20
21     factor_mean = F.mean(axis=0)
22     centered_factors = F - factor_mean
23     omega = centered_factors.T @ centered_factors / (T - 1)
24
25     # Stop if covariance matrices are not positive definite.
26     np.linalg.cholesky(sigma)
27     np.linalg.cholesky(omega)
28
29     alpha_term = alpha @ np.linalg.solve(sigma, alpha)
30     factor_term = factor_mean @ np.linalg.solve(omega, factor_mean)
31
32     statistic = ((T - N - K) / N) * alpha_term / (1 + factor_term)
33     p_value = f.sf(statistic, N, T - N - K)
34
35     return {
36         "T": T,
37         "N": N,
38         "K": K,
39         "GRS F-statistic": float(statistic),
40         "p-value": float(p_value),
41         "Residual denominator": T - K - 1,
42         "Factor denominator": T - 1,
43         "F df numerator": N,
44         "F df denominator": T - N - K
45     }
46
47     capm_grs = grs_test(
48         capm_results["Alpha"], capm_residuais, F_capm
49     )
50
51     print(pd.Series(capm_grs).to_string())

```



```

1     X_aug = sm.add_constant(F_aug, has_constant="add")
2
3     aug_rows = []
4     aug_residuais = pd.DataFrame(index=Y.index)
5     aug_fitted = pd.DataFrame(index=Y.index)
6
7     for portfolio in portfolio_cols:
8         model = sm.OLS(Y[portfolio], X_aug, missing="raise").fit()
9
10        aug_rows.append({
11            "Portfolio": portfolio,
12            "Alpha": model.params["const"],
13            "Alpha SE": model.bse["const"],
14            "Alpha t-stat": model.tvalues["const"],
15            "Market beta": model.params["RM-RF"],
16            "Market beta t-stat": model.tvalues["RM-RF"],
17            "RMW beta": model.params["RMW"],
18            "RMW beta t-stat": model.tvalues["RMW"]
19        })
20
21        aug_residuais[portfolio] = model.resid
22        aug_fitted[portfolio] = model.fittedvalues
23
24    aug_results = pd.DataFrame(aug_rows).set_index("Portfolio")
25    print(aug_results.to_string(float_format=lambda x: f"{x:.6f}"))

```



```

1     aug_grs = grs_test(
2         aug_results["Alpha"], aug_residuais, F_aug
3     )
4
5     print(pd.Series(aug_grs).to_string())

```

```

1 validation_passed = False
2 validation_rows = []
3 audit_rows = []
4
5 models = [
6     ("CAPM", capm_results, capm_residuals, capm_fitted,
7      F_capm, capm_grs, 1),
8     ("Market + RMW", aug_results, aug_residuals, aug_fitted,
9      F_aug, aug_grs, 2)
10 ]
11
12 for name, results, residuals, fitted, factors, result, expected_k in models:
13     alpha = results["Alpha"].to_numpy()
14     original_grs = result["GRS F-statistic"]
15
16     # Check 1: zero alpha.
17     zero_grs = grs_test(
18         np.zeros(len(alpha)), residuals, factors
19     )["GRS F-statistic"]
20
21     # Check 2: reverse the portfolio order consistently.
22     reorder_grs = grs_test(
23         alpha[::-1], residuals.iloc[:, ::-1], factors
24     )["GRS F-statistic"]
25
26     # Check 3: scale all returns and factors, then refit.
27     scaled_X = sm.add_constant(factors * 100, has_constant="add")
28     scaled_alpha = []
29     scaled_residuals = pd.DataFrame(index=Y.index)
30
31     for portfolio in portfolio_cols:
32         scaled_model = sm.OLS(Y[portfolio] * 100, scaled_X).fit()
33         scaled_alpha.append(scaled_model.params["const"])
34         scaled_residuals[portfolio] = scaled_model.resid
35
36     scale_grs = grs_test(
37         scaled_alpha, scaled_residuals, factors * 100
38     )["GRS F-statistic"]
39
40     checks = [
41         ("Zero alpha", zero_grs,
42          np.isclose(zero_grs, 0, rtol=0, atol=1e-12)),
43         ("Reorder invariance", reorder_grs,
44          np.isclose(reorder_grs, original_grs, rtol=1e-9, atol=1e-12)),
45         ("Scale invariance", scale_grs,
46          np.isclose(scale_grs, original_grs, rtol=1e-9, atol=1e-12))
47     ]
48
49     for check, value, passed in checks:
50         validation_rows.append({
51             "Model": name,
52             "Check": check,
53             "GRS": value,
54             "Result": "PASS" if passed else "FAIL"
55         })
56
57     T, N = residuals.shape
58     K = factors.shape[1]
59
60     # Verify residuals equal observed excess returns minus fitted returns.
61     residual_check = np.allclose(
62         residuals.to_numpy(),
63         (Y - fitted).to_numpy(),
64         rtol=1e-10, atol=1e-10
65     )
66
67     # OLS residuals should be orthogonal to the regressors.
68     X = sm.add_constant(factors, has_constant="add").to_numpy()

```

```

69     orthogonal_check = np.allclose(
70         X.T @ residuals.to_numpy(), 0, rtol=0, atol=1e-7
71     )
72
73     specification_check = (
74         K == expected_k
75         and result["K"] == expected_k
76         and result["N"] == 25
77         and result["Residual denominator"] == T - K - 1
78         and result["Factor denominator"] == T - 1
79         and result["F df numerator"] == N
80         and result["F df denominator"] == T - N - K
81     )
82
83     audit_rows.append({
84         "Model": name,
85         "K": K,
86         "Residual denominator": result["Residual denominator"],
87         "Factor denominator": result["Factor denominator"],
88         "F degrees of freedom": (N, T - N - K),
89         "OLS residual check": "PASS" if residual_check else "FAIL",
90         "Orthogonality": "PASS" if orthogonal_check else "FAIL",
91         "Specification": "PASS" if specification_check else "FAIL"
92     })
93
94     validation_results = pd.DataFrame(validation_rows)
95     grs_audit = pd.DataFrame(audit_rows)
96
97     print(validation_results.to_string(index=False))
98     print("\nGRS implementation audit:")
99     print(grs_audit.to_string(index=False))
100
101     assert validation_results["Result"].eq("PASS").all(), "GRS validation failed."
102     assert grs_audit["OLS residual check"].eq("PASS").all(), "Residual check failed."
103     assert grs_audit["Orthogonality"].eq("PASS").all(), "OLS orthogonality failed."
104     assert grs_audit["Specification"].eq("PASS").all(), "GRS specification failed."
105
106     validation_passed = True
107     print("\nAll checks passed. Continue to the comparison tables.")

```

Table 14: GRS validation, capstone implementation, 1963-07 to 2026-06 ($T = 756$).

Model	Check	GRS	Result
CAPM	Zero alpha	0.000000	PASS
CAPM	Reorder invariance	4.164336	PASS
CAPM	Scale invariance	4.164336	PASS
Market + RMW	Zero alpha	0.000000	PASS
Market + RMW	Reorder invariance	3.723390	PASS
Market + RMW	Scale invariance	3.723390	PASS

Residual covariance denominators: 754 (CAPM) and 753 (Market + RMW); factor covariance denominator 755; $F(25, 730)$ and $F(25, 729)$.

```

1  assert validation_passed, "Run the validation cell first."
2
3  comparison = pd.DataFrame([
4      {
5          "Model": "CAPM",
6          "Factors": "RM-RF",
7          "T": capm_grs["T"],
8          "N": capm_grs["N"],
9          "K": capm_grs["K"],
10         "GRS F-statistic": capm_grs["GRS F-statistic"],
11         "p-value": capm_grs["p-value"],

```

```

12     "Average RMW return": np.nan
13 },
14 {
15     "Model": "Augmented model",
16     "Factors": "RM-RF + RMW",
17     "T": aug_grs["T"],
18     "N": aug_grs["N"],
19     "K": aug_grs["K"],
20     "GRS F-statistic": aug_grs["GRS F-statistic"],
21     "p-value": aug_grs["p-value"],
22     "Average RMW return": common["RMW"].mean()
23 }
24 ])
25
26 assert capm_grs["GRS F-statistic"] > 0
27
28 grs_reduction = (
29     (capm_grs["GRS F-statistic"] - aug_grs["GRS F-statistic"])
30     / capm_grs["GRS F-statistic"] * 100
31 )
32
33 print(comparison.to_string(
34     index=False, na_rep="N/A",
35     formatters={
36         "GRS F-statistic": "{:.6f}".format,
37         "p-value": "{:.6e}".format,
38         "Average RMW return": "{:.6f}".format
39     }
40 ))
41 print(f"\nGRS percentage reduction: {grs_reduction:.6f}%")

1  assert validation_passed, "Run the validation cell first."
2
3  alpha_comparison = pd.DataFrame({
4      "Portfolio": portfolio_cols,
5      "CAPM alpha": capm_results["Alpha"].to_numpy(),
6      "CAPM alpha t-stat": capm_results["Alpha t-stat"].to_numpy(),
7      "Market + RMW alpha": aug_results["Alpha"].to_numpy(),
8      "Market + RMW alpha t-stat": aug_results["Alpha t-stat"].to_numpy(),
9      "RMW beta": aug_results["RMW beta"].to_numpy(),
10     "RMW beta t-stat": aug_results["RMW beta t-stat"].to_numpy()
11 })
12
13 alpha_comparison["Absolute alpha reduction"] = (
14     alpha_comparison["CAPM alpha"].abs()
15     - alpha_comparison["Market + RMW alpha"].abs()
16 )
17
18 alpha_sorted = alpha_comparison.sort_values(
19     "Absolute alpha reduction", ascending=False
20 ).copy()
21
22 print("Original portfolio order:")
23 print(alpha_comparison.to_string(
24     index=False, float_format=lambda x: f"{x:.6f}"
25 ))
26
27 print("\nSorted by absolute alpha reduction:")
28 print(alpha_sorted.to_string(
29     index=False, float_format=lambda x: f"{x:.6f}"
30 ))

1  assert validation_passed, "Run the validation cell first."
2
3  T = len(common)
4  rmw_mean = common["RMW"].mean()
5  rmw_sd = common["RMW"].std(ddof=1)

```

```

6  rmw_se = rmw_sd / np.sqrt(T)
7
8  assert rmw_se > 0
9  rmw_t = rmw_mean / rmw_se
10
11 rmw_mean_test = pd.DataFrame([
12     "T": T,
13     "Mean monthly RMW (%)": rmw_mean,
14     "Standard deviation (%)": rmw_sd,
15     "Standard error (%)": rmw_se,
16     "Mean t-statistic": rmw_t
17 ])
18
19 print("One-sample test: H0: E[RMW] = 0")
20 print(rmw_mean_test.to_string(
21     index=False, float_format=lambda x: f"{x:.6f}"
22 ))
23 print("\nThis tests the RMW mean, not the joint portfolio pricing errors.")

1  from scipy.stats import t as t_dist
2
3  return_sets = {
4     "CAPM factors": F_capm,
5     "Augmented factors": F_aug,
6     "CAPM factors + portfolios": pd.concat([F_capm, Y], axis=1),
7     "Augmented factors + portfolios": pd.concat([F_aug, Y], axis=1)
8  }
9
10 sr2 = {}
11
12 for name, returns in return_sets.items():
13     mean = returns.mean().to_numpy()
14     covariance = returns.cov().to_numpy()
15     sr2[name] = float(mean @ np.linalg.solve(covariance, mean))
16
17 capm_gain = sr2["CAPM factors + portfolios"] - sr2["CAPM factors"]
18 aug_gain = sr2["Augmented factors + portfolios"] - sr2["Augmented factors"]
19 gain_reduction = (1 - aug_gain / capm_gain) * 100
20
21 print("SHARPE-RATIO RESULTS (monthly)")
22 for name, value in sr2.items():
23     print(f"{name}: SR = {np.sqrt(value):.4f}, SR2 = {value:.6f}")
24
25 print(f"\nCAPM incremental SR2: {capm_gain:.6f}")
26 print(f"Augmented incremental SR2: {aug_gain:.6f}")
27 print(f"Incremental SR2 reduction: {gain_reduction:.6f}%")
28 print(f"CAPM scaling denominator: {1 + sr2['CAPM factors']:.4f}")
29 print(f"Augmented scaling denominator: {1 + sr2['Augmented factors']:.4f}")
30
31 rmw_p = 2 * t_dist.sf(abs(rmw_t), df=len(common) - 1)
32
33 print("\nFACTOR MEANS AND RMW MEAN TEST")
34 print(f"Mean RM-RF: {common['RM-RF'].mean():.6f}% per month")
35 print(f"Mean RMW: {rmw_mean:.6f}% per month")
36 print(f"RMW mean t-statistic: {rmw_t:.6f}")
37 print(f"RMW mean-test p-value: {rmw_p:.6f}")
38
39 capm_abs = capm_results["Alpha"].abs()
40 aug_abs = aug_results["Alpha"].abs()
41 alpha_reduction = capm_abs - aug_abs
42
43 print("\nABSOLUTE-ALPHA SUMMARY (monthly percentage points)")
44 print(f"CAPM total absolute alpha: {capm_abs.sum():.6f}")
45 print(f"Augmented total absolute alpha: {aug_abs.sum():.6f}")
46 print(f"Portfolios with lower absolute alpha: {(alpha_reduction > 0).sum()} of 25")

1  assert validation_passed, "Run the validation cell first."

```

```

2
3 print("COMMON SAMPLE")
4 print("Start:", common.index.min())
5 print("End:", common.index.max())
6 print("T:", len(common))
7
8 print("\nGRS RESULTS")
9 print(
10     f"CAPM: F = {capm_grs['GRS F-statistic']:.6f}, "
11     f"p = {capm_grs['p-value']:.6e}"
12 )
13 print(
14     f"Market + RMW: F = {aug_grs['GRS F-statistic']:.6f}, "
15     f"p = {aug_grs['p-value']:.6e}"
16 )
17 print(f"GRS percentage reduction: {grs_reduction:.6f}%")
18
19 print("\nRMW MEAN TEST")
20 print(rmw_mean_test.to_string(
21     index=False, float_format=lambda x: f"{x:.6f}"
22 ))
23
24 print("\nMODEL COMPARISON")
25 print(comparison.to_string(
26     index=False, na_rep="N/A",
27     formatters={
28         "GRS F-statistic": "{:.6f}".format,
29         "p-value": "{:.6e}".format,
30         "Average RMW return": "{:.6f}".format
31     }
32 ))
33
34 print("\nALPHA COMPARISON: ORIGINAL ORDER")
35 print(alpha_comparison.to_string(
36     index=False, float_format=lambda x: f"{x:.6f}"
37 ))
38
39 print("\nALPHA COMPARISON: SORTED COPY")
40 print(alpha_sorted.to_string(
41     index=False, float_format=lambda x: f"{x:.6f}"
42 ))
43
44 print("\nGRS VALIDATION")
45 print(validation_results.to_string(index=False))
46 print(grs_audit.to_string(index=False))
47
48 print("\nTECHNICAL NOTE")
49 print(
50     "The implementation uses one common sample, workbook RM-RF and RF, "
51     "and only RMW from the external CSV. RF is subtracted once from "
52     "raw portfolio returns. GRS follows the assignment's displayed "
53     "formula, using OLS residual covariance divided by T-K-1, factor "
54     "covariance divided by T-1, and F(N, T-N-K) degrees of freedom."
55 )
56 print(
57     "Special attention was required for the workbook's multiple header "
58     "rows, YYYYMM dates, and the CSV's separate annual section. "
59     "The portfolio labels preserve the original workbook ordering."
60 )
61 print(
62     "Date alignment excludes 444 workbook months before July 1963 "
63     "and two RMW months after June 2026 in these uploaded files. "
64     "No missing observations remain in the 756-month common sample. "
65 )
66
67 pd.set_option("display.max_rows", None)
68 pd.set_option("display.max_columns", None)
69 pd.set_option("display.width", None)

```

```

4 pd.set_option("display.max_colwidth", None)
5
6 print(alpha_comparison.to_string(
7     index=False,
8     float_format=lambda x: f"{x:.6f}"
9 ))

```

Table 15: Alpha comparison, CAPM vs. Market + RMW, 25 size and BE/ME portfolios, monthly percentage points. S1 = smallest size quintile, BM1 = lowest BE/ME.

Portfolio	CAPM		Market + RMW		RMW loading		$ \hat{\alpha}_{CAPM} - \hat{\alpha}_{aug} $
	$\hat{\alpha}$	t	$\hat{\alpha}$	t	$\hat{\beta}_{RMW}$	t	
S1_BM1	-0.522	-2.93	-0.212	-1.33	-1.029	-14.55	0.310
S1_BM2	0.059	0.38	0.328	2.32	-0.891	-14.17	-0.268
S1_BM3	0.137	1.06	0.296	2.40	-0.527	-9.60	-0.159
S1_BM4	0.347	2.64	0.487	3.84	-0.465	-8.23	-0.140
S1_BM5	0.473	3.16	0.598	4.06	-0.416	-6.33	-0.125
S2_BM1	-0.266	-1.96	-0.070	-0.55	-0.651	-11.58	0.196
S2_BM2	0.103	0.92	0.213	1.96	-0.368	-7.59	-0.111
S2_BM3	0.245	2.37	0.285	2.74	-0.132	-2.85	-0.040
S2_BM4	0.291	2.75	0.344	3.24	-0.174	-3.68	-0.052
S2_BM5	0.320	2.43	0.405	3.10	-0.283	-4.86	-0.085
S3_BM1	-0.220	-2.01	-0.089	-0.85	-0.434	-9.34	0.131
S3_BM2	0.124	1.43	0.158	1.81	-0.110	-2.83	-0.033
S3_BM3	0.144	1.69	0.140	1.62	0.014	0.36	0.004
S3_BM4	0.272	2.85	0.276	2.86	-0.013	-0.29	-0.004
S3_BM5	0.344	2.76	0.359	2.85	-0.050	-0.89	-0.015
S4_BM1	-0.069	-0.83	0.020	0.25	-0.297	-8.24	0.049
S4_BM2	0.016	0.23	-0.002	-0.02	0.058	1.90	0.014
S4_BM3	0.143	1.81	0.109	1.38	0.113	3.20	0.034
S4_BM4	0.276	3.04	0.270	2.94	0.022	0.53	0.007
S4_BM5	0.221	1.84	0.236	1.95	-0.049	-0.92	-0.015
S5_BM1	0.020	0.33	-0.049	-0.87	0.229	9.11	-0.030
S5_BM2	0.030	0.51	-0.031	-0.56	0.201	8.09	-0.002
S5_BM3	0.091	1.23	0.054	0.72	0.125	3.77	0.038
S5_BM4	0.021	0.22	-0.025	-0.26	0.152	3.57	-0.004
S5_BM5	0.194	1.47	0.234	1.76	-0.133	-2.24	-0.040

3a report

What the code got right. The code correctly constructed portfolio excess returns, used the same 1963-07 to 2026-06 common sample for both models, and estimated the CAPM with ($K = 1$) and the augmented model with ($K = 2$). The GRS implementation used the residual covariance matrix with denominator $(T - K - 1)$, the factor covariance matrix with denominator $(T - 1)$, the correct scaling term, and $(F(N, T - N - K))$ degrees of freedom. The zero-alpha, reorder-invariance, and scale-invariance checks also passed, and the independently recomputed GRS statistics matched the notebook.

What we corrected or added. We did not need to correct the regression or GRS calculations themselves. The main correction was to the alpha-table display: the original notebook output was visually truncated, so we added a separate cell that prints all 25 portfolios for Step 3(b). Additional Sharpe-ratio and summary statistics were added later for Step 4, rather than to fix an error in the Step 3 implementation.

Step 3b

LLM's Explanation. Adding RMW helps price a relatively small subset of portfolios, especially the low-BE/ME portfolios. The largest improvements are for S1_BM1 (absolute alpha falls by 0.310), S2_BM1 (0.196), and S3_BM1 (0.131). Smaller improvements occur for portfolios such as S4_BM1, S4_BM3, and S5_BM3. RMW does not help many of the remaining portfolios. The largest deteriorations are S1_BM2 (−0.268), S1_BM3 (−0.159), S1_BM4 (−0.140), and S1_BM5 (−0.125). Overall, RMW appears most useful for some low-BE/ME portfolios, particularly among smaller firms, but it does not consistently reduce pricing errors across the full set of 25 portfolios.

Critical Assessing. The LLM's portfolio-level figures are accurate, and it correctly identifies the largest reductions (S1_BM1, S2_BM1, S3_BM1) and deteriorations (S1_BM2–S1_BM5) in absolute alpha. However, it understates how limited the improvement is: absolute alphas fall for only 9 of 25 portfolios, several of those reductions are below 0.015, and total absolute alpha rises from 4.95 to 5.29. Its claim that RMW helps “particularly among smaller firms” is misleading, because the smallest-size quintile also contains the four largest deteriorations. Most importantly, it misses the pattern that organizes the table: given RMW's positive sample mean, pricing errors shrink only where the sign of the RMW loading offsets the sign of the CAPM alpha, and grow where the loading pushes an already-positive alpha further from zero, as in small value portfolios. The table therefore only partly supports our hypothesis: weakly exposed portfolios change little, but strong RMW exposure is associated with large alpha changes in either direction, not necessarily reductions.

Step 4

Table 16: GRS tests, CAPM vs. augmented model, 25 size and BE/ME portfolios, 1963-07 to 2026-06 ($T = 756$).

Model	Factors	GRS F-statistic		p -value	Mean RM-RF	Mean RMW
CAPM	RM-RF	4.164336	9.853287×10^{-11}		0.600794%	—
Augmented	RM-RF + RMW	3.723390	4.213838×10^{-9}		0.600794%	0.242804%

4(a) Adding RMW lowers GRS from 4.164336 to 3.723390, a 10.588612% reduction. This indicates improvement in the joint pricing-error criterion, but the reduction itself is not a formal test of statistically significant improvement. The augmented-model p -value remains 4.213838×10^{-9} , strongly rejecting the joint null that all 25 alphas equal zero.

Using the same uploaded sample and sample covariances with denominator $T - 1$, the incremental maximum squared monthly Sharpe ratio from adding the test assets falls from 0.145390 to 0.132731, approximately 8.707282%. Thus, the test assets offer a smaller remaining Sharpe-ratio improvement, but no combination of RM-RF and RMW is mean-variance efficient relative to the factors plus the 25 test assets. Most of the change comes from the factor side: the maximum monthly Sharpe ratio of the factors rises from 0.1346 (market alone) to 0.1917 (market + RMW), while that of the factors plus test assets rises only from 0.4044 to 0.4117. The percentage differs from the GRS reduction mainly because GRS divides this increment by $1 + SR_f^2$, which rises from 1.0181 to 1.0367 when RMW is added; the remainder reflects the degrees-of-freedom and covariance-denominator adjustments.

4(b) Defining improvement as $|\alpha_{CAPM}| - |\alpha_{augmented}|$, the largest reductions occur in S1_BM1 (0.310156), S2_BM1 (0.196118), and S3_BM1 (0.130670), measured in monthly percentage points. These are the lowest-BE/ME portfolios in the first three size quintiles. However, absolute alphas

decline for only 9 of 25 portfolios. The largest deteriorations occur in S1_BM2 (-0.268286), S1_BM3 (-0.158904), S1_BM4 (-0.140086), and S1_BM5 (-0.125237). Stronger absolute RMW exposure therefore does not consistently imply improvement: S1_BM2 has a large loading of -0.890504 , yet its positive alpha increases substantially.

The hypothesis is partially supported. GRS falls and pricing errors remain, as predicted, and the three largest reductions occur in portfolios with strong negative RMW loadings (-1.029 , -0.651 , -0.434). Weakly exposed portfolios ($|\beta_{RMW}| < 0.06$) change by at most 0.015 , also consistent with the hypothesis. However, strong exposure is associated with large alpha changes in either direction rather than consistently larger reductions: of the seven portfolios with $|\beta_{RMW}| \geq 0.4$, three improve and four deteriorate.

4(c)

1. RMW's mean return: The average is 0.242804% per month, with $t = 2.966978$. A conventional two-sided t -test rejects a zero mean at the 1% level ($p \approx 0.003102$).
2. Pricing-error improvement: GRS decreases by 10.588612% , indicating a modest improvement in the joint criterion. Nevertheless, the augmented-model p -value of 4.213838×10^{-9} and the mixed portfolio-level alpha changes show that substantial pricing errors remain.

These tests address different questions. The mean-return test asks whether RMW's average return, its implied premium in the time-series model, differs from zero; GRS tests whether the 25 intercepts are jointly zero, accounting for residual covariances and factor moments. A factor with an imprecisely estimated mean can still reduce pricing errors, because each portfolio's alpha change depends on its RMW loading and the point estimate of the mean, not on the precision of that estimate. Conversely, RMW's significant mean here does not imply that it adequately prices the test assets.

4(d) The main limitation is overlap with the evidence used to establish RMW. Fama and French (2015) documented the five-factor model over July 1963–December 2013 and evaluated it on the same 25 size-B/M portfolios, so about 80% of this test's 756 months (606), and the test assets themselves, overlap the factor's discovery sample; only January 2014–June 2026 (150 months) postdates it. Factor construction adds further dependence: RMW and the test portfolios are formed from the same U.S. CRSP/Compustat universe using NYSE-based breakpoints, so they share underlying stocks. This is therefore not an independent out-of-sample test, and the historical GRS reduction may overstate performance in new data; this limits external validity without invalidating the in-sample result.

Prompt 1 — Step 3(a): Implementation. “Write Python code to estimate the CAPM and augmented RM-RF + RMW time-series regressions for the 25 Size/BE/ME portfolios and run the GRS test. Use the same common sample for both models, report the GRS statistics and p -values, create an alpha comparison table, and validate the GRS implementation.”

Prompt 2 — Step 3(b): Interpretation. “Using the 25-portfolio alpha comparison table, identify which portfolios RMW helps price and which it does not. Focus on changes in absolute alpha, patterns across size and BE/ME, and whether the results are consistent with my Step 2 hypothesis.”

Step 5

Table 17: AI interaction Log.

Interaction	Prompt/Purpose	What the LLM produced correctly	What required correction or verification
Step 3(a): Implementation	Prompt 1 above: implement and validate CAPM and RM-RF + RMW regressions and GRS tests.	The final implementation correctly handled the two-factor regression and the assignment's GRS framework, producing GRS statistics, p -values, and portfolio alphas.	We verified the common sample, excess-return construction, $K = 1$ versus $K = 2$, covariance denominators, degrees of freedom, and zero-alpha, reorder-invariance, and scale-invariance checks. No calculation correction was needed; we expanded the truncated alpha-table display to show all 25 portfolios and added a cell computing the Sharpe-ratio quantities, RMW mean-test p -value, mean RM-RF, and absolute-alpha totals.
Step 3(b): Alpha interpretation	Prompt 2 above: interpret absolute-alpha changes, size/BE/ME patterns, and consistency with our hypothesis.	All quoted figures were accurate, including the largest improvements in S1_BM1–S3_BM1 and deteriorations in S1_BM2–S1_BM5.	The description needed precision: only 9 of 25 portfolios improved, and “particularly among smaller firms” obscured the four largest deteriorations within the smallest-size quintile. The response omitted the hypothesis comparison and the role of signed RMW exposures in explaining alpha changes; significant exposure alone would not establish improvement.

Reflection. Working with the LLM clarified that GRS jointly evaluates regression intercepts using the residual covariance matrix, accounting for correlations among pricing errors. Because GRS evaluates the alphas using the inverse residual covariance matrix, its statistic can fall even when most individual absolute alphas rise, as in our results. Verifying covariance denominators and degrees of freedom showed us why invariance checks alone cannot establish that an implementation is correct. We now understand GRS as a test of the factors' mean-variance efficiency relative to the test assets, rather than a count of individually significant alphas.